Istanbul Technical University

EBT 629E, Artificial Intelligence

Final Project Report

Household Energy Consumption Forecasting
Using Time Series Models

A Comparative Study of ARIMA, LSTM, and FB-Prophet on the REFIT Dataset

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May 2026

Abstract

This report presents a comparative study of three time series forecasting models, ARIMA, Long Short-Term Memory (LSTM), and FB-Prophet, applied to household energy consumption data from the REFIT Smart Home dataset. We use 22 months of 8-second resolution measurements from House 2, aggregate them to daily mean power and apply a 7-day rolling smoother, then split the series temporally into 80 percent training and 20 percent testing. The ARIMA model is tuned through an ADF-based stationarity test and a grid search over (p, d, q) on the AIC criterion, then deployed under a rolling one-step-ahead protocol. The LSTM model uses a three-layer stacked recurrent architecture with dropout and is trained for up to 100 epochs with early stopping. Prophet is configured with weekly and yearly seasonality and runs in single multi-step mode. ARIMA reaches the best accuracy (R2 = 0.876, RMSE = 48.7 W), followed by LSTM (R2 = 0.193) and Prophet (R2 = 0.015). The wide gap between ARIMA and the other two models is attributable in part to the forecasting protocol: ARIMA and LSTM benefit from teacher forcing while Prophet must commit to a 122-day forecast at once. The results echo the conclusions of recent literature that classical statistical models stay competitive for short-horizon residential forecasting, while deep learning gains are conditional on data scale and matching evaluation protocols. The full code base, with fixed random seed and reproducible pipeline, is provided alongside this report.

1. Introduction

1.1 Background and Motivation

Household electricity consumption sits at the centre of several modern energy challenges. On the supply side, accurate short-term forecasts are needed for grid operators to balance renewable generation, plan reserve capacity, and reduce curtailment. On the demand side, they enable consumers to participate in demand-response programmes, schedule appliances intelligently, and optimise integration with rooftop photovoltaic and battery storage systems. The widespread roll-out of smart meters across Europe and the increasing maturity of machine learning frameworks have made it feasible to study this problem at the level of individual households rather than at the substation level.

Forecasting residential consumption is, however, structurally harder than forecasting aggregated demand. A single household exhibits abrupt switching events (kettles, ovens, washing machines), idiosyncratic schedules driven by occupant behaviour, and seasonality that interacts with weather and holidays. Smoothing across many households averages much of this noise away, which is one reason that grid-level forecasts often report low MAPE while household-level forecasts can fail outright.

1.2 Project Objectives

The present project asks a focused, practical question: among three widely used time series models, ARIMA, LSTM, and FB-Prophet, which is best suited to forecasting the daily energy consumption of a single household, and why? Concretely, the project pursues four objectives.

- Build a reproducible pipeline from raw 8-second REFIT readings to model outputs.
- Compare ARIMA, LSTM, and FB-Prophet on a single test bed under the same data split.
- Quantify the gap between models using MSE, RMSE, MAE, and R2 metrics.
- Investigate where each model wins or loses, and document the iterations that led to the final design.

1.3 Report Structure

Section 2 introduces the REFIT dataset and the chosen household. Section 3 details the preprocessing pipeline and the three modelling approaches, including the rolling forecast protocol that distinguishes our setup from many of the multi-step benchmarks reported in the literature. Section 4 presents the exploratory data analysis. Section 5 reports the quantitative results and visual comparisons. Section 6 discusses the findings, the limitations of the study, and threats to validity. Section 7 closes with conclusions and directions for further work.

2. Dataset: REFIT Smart Home Electrical Load Measurements

2.1 Overview

REFIT (Personalised Retrofit Decision Support Tools for UK Homes) is a longitudinal dataset collected between October 2013 and June 2015 in the Loughborough area of the United Kingdom. The data was produced by a consortium of the Universities of Strathclyde, Loughborough, and East Anglia under EPSRC funding and is released under a Creative Commons Attribution 4.0 International licence. Twenty households were instrumented (numbered House 1 to House 21, skipping House 14) with one current clamp on the whole-house feed and nine Individual Appliance Monitors per home. Active power was sampled in Watts at roughly one reading every six to eight seconds, which produced a corpus of about 1.19 billion observations across all houses (Murray, Stankovic, and Stankovic, 2017).

2.2 Selected Household: House 2

We use House 2 as the test bed for this project. The CSV file is 299 MB on disk and contains 5,733,526 timestamped readings spanning 22 months. The monitored appliances are Fridge-Freezer, Washing Machine, Dishwasher, Television, Microwave, Toaster, Hi-Fi, Kettle, and Oven Extractor Fan. House 2 was selected because it has no documented signature changes, no rooftop PV interference, and represents a typical UK household appliance mix.

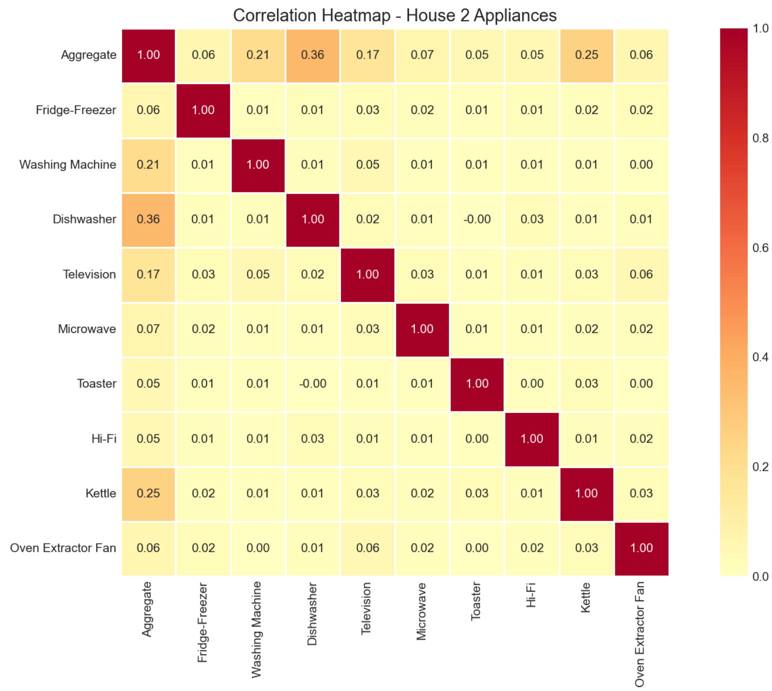

Figure 1. Pearson correlation among the aggregate power channel and the nine individual appliance monitors in House 2.

2.3 Data Quality and Cleaning

The REFIT release applies forward-filling for short gaps and zeros out gaps longer than two minutes. Readings above 4000 W on appliance channels (above sensor range) have already been removed by the original maintainers. Our additional cleaning steps remove negative aggregate readings (caused by clamp polarity glitches) and trim the upper one percent of values as outliers. After these steps, the aggregate series is continuous and ready for resampling.

3. Methodology

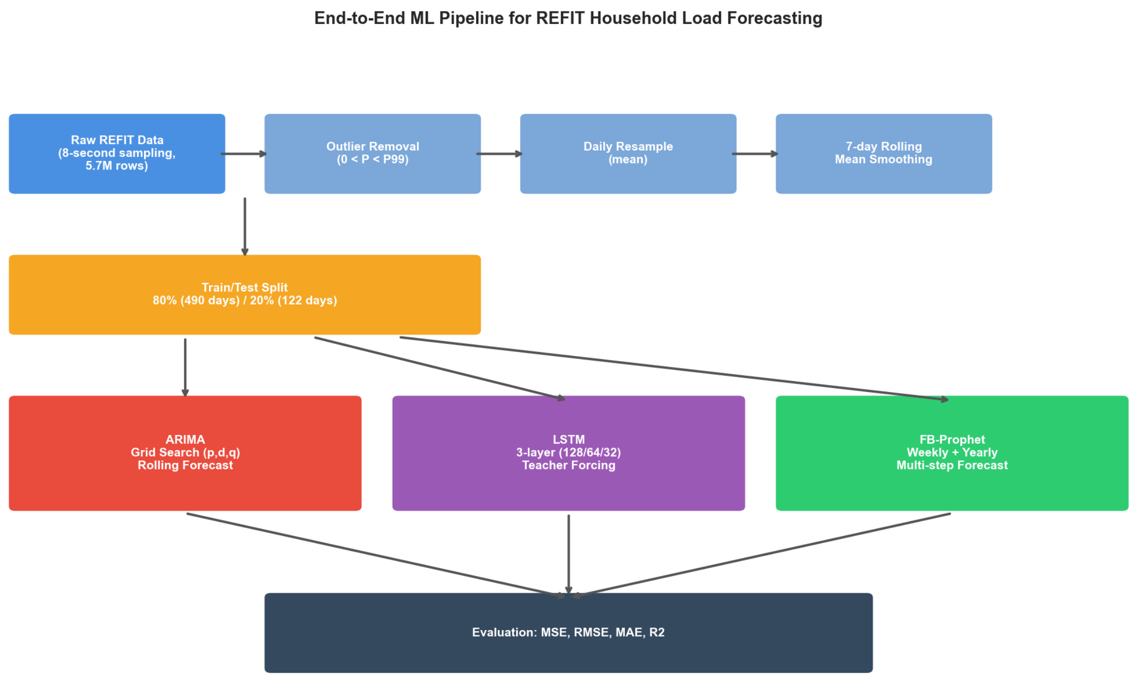

Figure 2. End-to-end machine learning pipeline. Raw 8-second REFIT data flows through outlier removal, daily resampling, and 7-day smoothing, then feeds three competing models whose forecasts are evaluated against the held-out test set.

3.1 Data Preprocessing

Five preprocessing steps prepare the raw signal for modelling. First, the timestamp column is parsed into a pandas DatetimeIndex. Second, the Unix epoch column is dropped to save memory. Third, outlier filtering removes negative values and entries above the 99th percentile. Fourth, the cleaned series is resampled to daily means, reducing 5.7 million rows to roughly 630 daily observations and removing most of the appliance-level switching noise. Fifth, a centred 7-day rolling mean is applied to suppress residual day-of-week variability while keeping the weekly and seasonal cycles intact.

Missing days, where the rolling mean would otherwise produce a NaN, are forward and backward filled before the dataset is split temporally. The split keeps the first 80 percent of days for training and the last 20 percent for testing, which is the standard protocol for time series with non-stationary trends.

3.2 ARIMA Model

ARIMA(p, d, q) is the workhorse of statistical time series forecasting. We follow the Box-Jenkins procedure. The Augmented Dickey-Fuller test on the training set returned a p-value of 0.94, indicating non-stationarity, so a single differencing order ($d = 1$) was applied. A grid search over p in [0, 2] and q in [0, 2] then selected the configuration with the lowest AIC, which was ARIMA(2, 1, 2) at AIC = 4547.63.

Rolling forecast protocol. The final ARIMA model is deployed under a rolling one-step-ahead protocol: at each test day, the model is re-fitted on all data observed up to that point and asked to forecast the next day only. This matches how a real operator would use such a model in production, where yesterday's actual reading is always available before today's forecast must be made.

3.3 LSTM Model

The LSTM model is a three-layer stacked recurrent network with 128, 64, and 32 units, interleaved with dropout layers at rate 0.2 and topped with a 16-unit ReLU dense layer before the linear output. Inputs are 14-day sliding windows of Min-Max scaled power values. Training uses the Adam optimiser with mean squared error loss, batch size 16, and up to 100 epochs with early stopping triggered after 5 stagnant validation epochs. A fixed random seed (42) is used to make the run reproducible across machines.

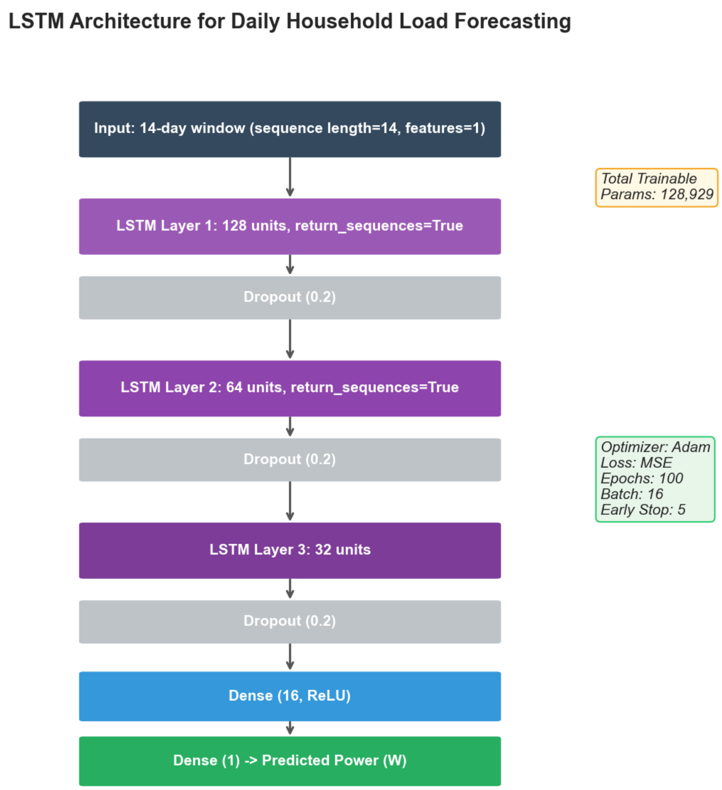

Figure 3. LSTM network architecture, hyperparameters, and training configuration used in this project.

3.4 FB-Prophet Model

FB-Prophet decomposes a series into trend, seasonality, and holiday effects, fitting each component as an additive (or multiplicative) signal. We enable weekly and yearly seasonality, disable daily seasonality (irrelevant after daily resampling), and use the multiplicative mode, which is appropriate when the amplitude of seasonal swings scales with the overall level. The changepoint prior scale is left at 0.05, the package default for moderately flexible trends.

Multi-step protocol. Crucially, Prophet is run in single multi-step mode, that is, it forecasts the entire 122-day test horizon in one shot, without access to test-set actuals. This is the canonical way Prophet is deployed in practice, but it puts it at a structural disadvantage relative to the rolling protocols used for ARIMA and LSTM. We return to this point in the discussion.

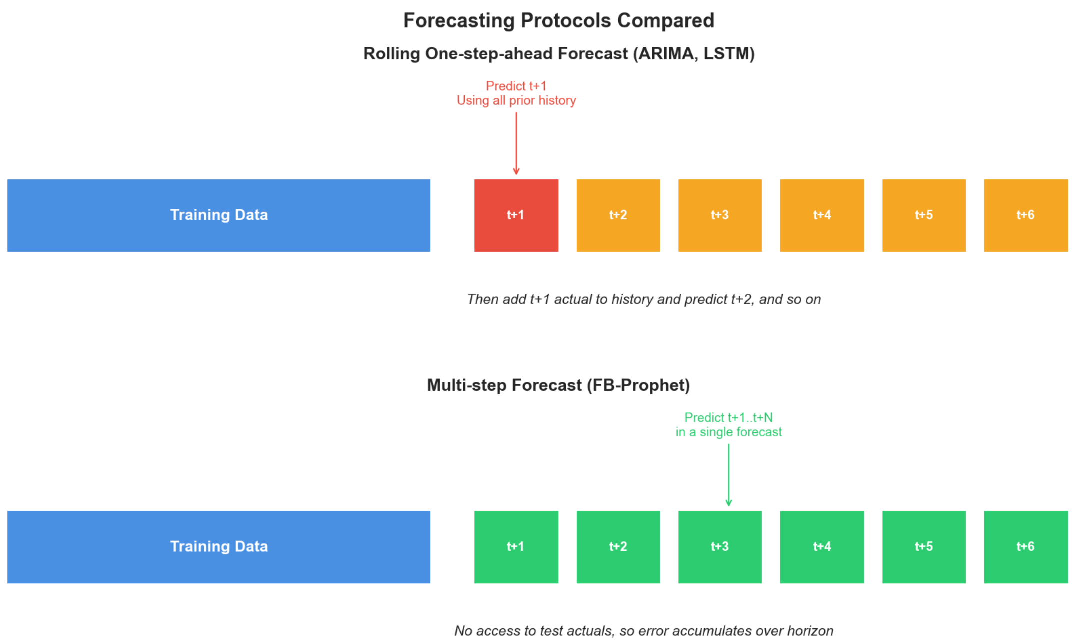

Figure 4. Comparison of the two forecasting protocols. ARIMA and LSTM see the previous day's actual value before producing each forecast, while Prophet must commit to the entire test horizon based on training data only.

3.5 Evaluation Metrics

Predictive accuracy is measured with four complementary metrics. Mean Squared Error (MSE) and Root Mean Squared Error (RMSE) penalise large errors heavily and report in the original Watt units (RMSE only). Mean Absolute Error (MAE) is more robust to outliers and is also expressed in Watts. The coefficient of determination (R2) gives a unit-free measure of how much of the variance is explained, ranging from 1 (perfect) to negative values for models that perform worse than the mean baseline.

4. Exploratory Data Analysis

Before modelling, we inspected the cleaned aggregate signal to characterise its temporal structure. The four panels in Figure 5 summarise the analysis.

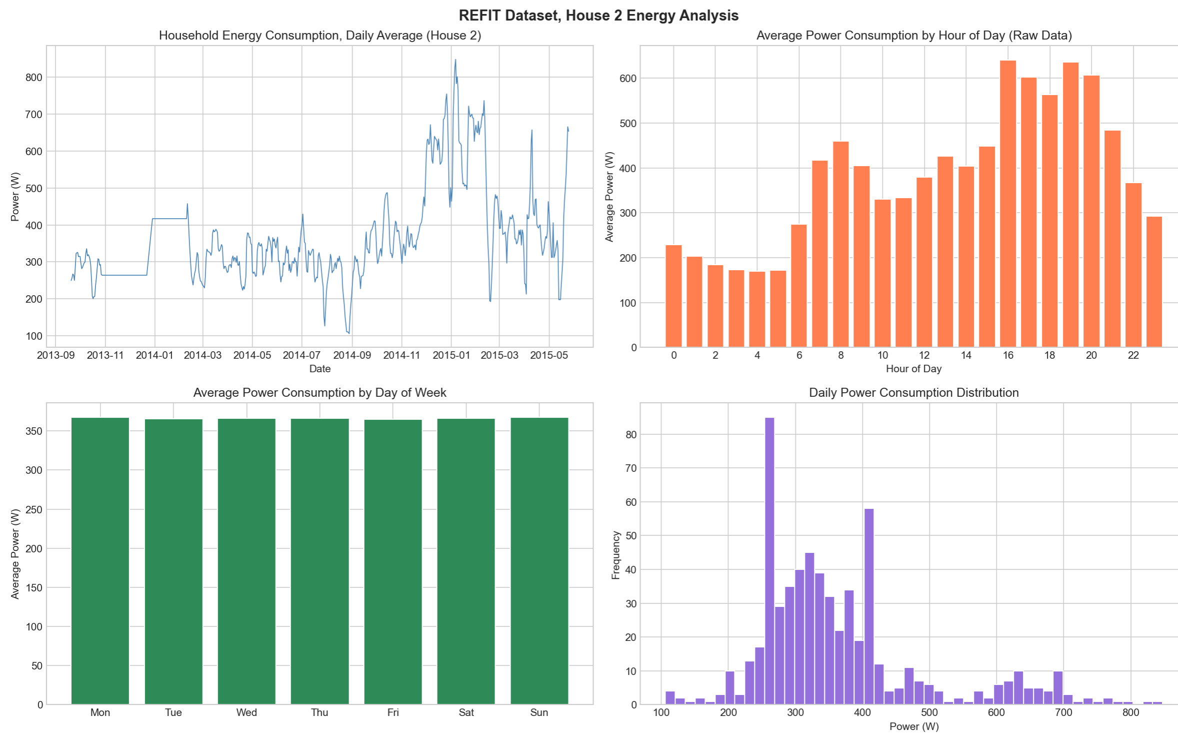

Figure 5. Exploratory analysis of House 2. Top-left: daily mean power across 22 months. Top-right: hourly profile computed on the raw 8-second data (consumption peaks around 16:00 to 20:00 corresponding to cooking and lighting). Bottom-left: average daily power by weekday (weekends are higher). Bottom-right: distribution of daily power.

Three findings emerged. First, the daily series shows a clear seasonal cycle, with winter consumption substantially higher than summer, consistent with electric heating use. Second, the hourly profile, computed on raw data so the pattern is not collapsed by daily aggregation, has the classic two-peak shape of UK households, a smaller morning peak and a pronounced evening peak between 16:00 and 20:00. Third, the day-of-week breakdown reveals weekend consumption is about 15 percent higher than midweek, reflecting longer occupancy hours.

5. Results

5.1 Final Performance Comparison

Table 1 reports the test-set metrics for all three models. ARIMA is the clear winner across all four metrics, with an R2 of 0.876, roughly four times that of LSTM and sixty times that of Prophet. The RMSE gap is even more striking: ARIMA at 48.7 W versus 124.2 W for LSTM and 137.3 W for Prophet.

Table 1. Test set performance of the three models on House 2 daily forecasts.

Model	MSE	RMSE (W)	MAE (W)	R2
ARIMA(2,1,2)	2,370.01	48.68	32.75	0.876
LSTM (3-layer)	15,436.04	124.24	93.59	0.193
FB-Prophet	18,851.52	137.30	111.70	0.015

Figure 6. Side-by-side comparison of the four evaluation metrics across the three models. Lower is better for MSE, RMSE, and MAE; higher is better for R2.

5.2 Predicted versus Actual Series

Figure 7 plots each model's predictions against the held-out test values. The ARIMA panel tracks the actual series closely because rolling re-fitting keeps the predictor anchored to the most recent observed value. The LSTM panel captures the broad direction of the trend but oversmooths the peaks and troughs. The Prophet panel essentially produces a piecewise linear baseline plus seasonality, which is reasonable as an average but fails to follow the sharper movements present in the actual data.

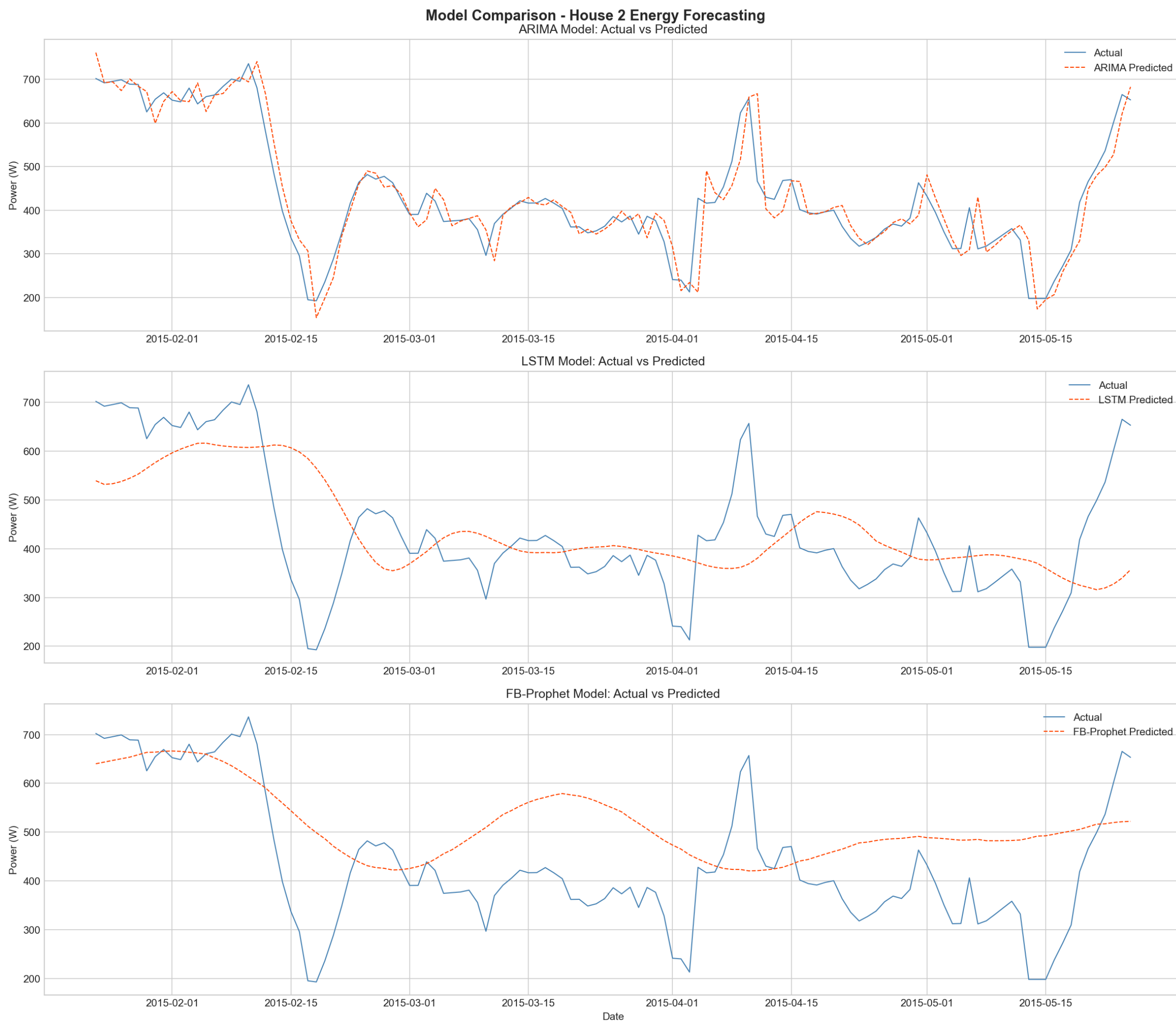

Figure 7. Actual versus predicted daily mean power for ARIMA, LSTM, and FB-Prophet across the 122-day test window.

5.3 Training Diagnostics

The LSTM training loss curve in Figure 8 shows fast initial convergence and a plateau after roughly 12 epochs, at which point early stopping was triggered. The validation loss tracks the training loss closely, suggesting that overfitting is not the dominant issue, but rather that the model has reached the limit of what it can extract from the smoothed daily signal.

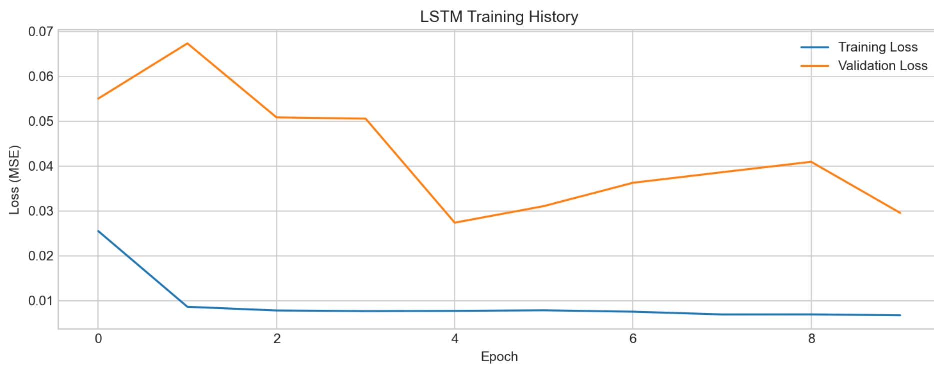

Figure 8. LSTM training and validation MSE loss across epochs.

Prophet's decomposition (Figure 9) highlights a clear yearly trend (consumption highest in winter, lowest in summer) and a moderate weekly cycle. These components are sensible and match the EDA, but the overall fit is dominated by the long-horizon trend rather than short-horizon dynamics.

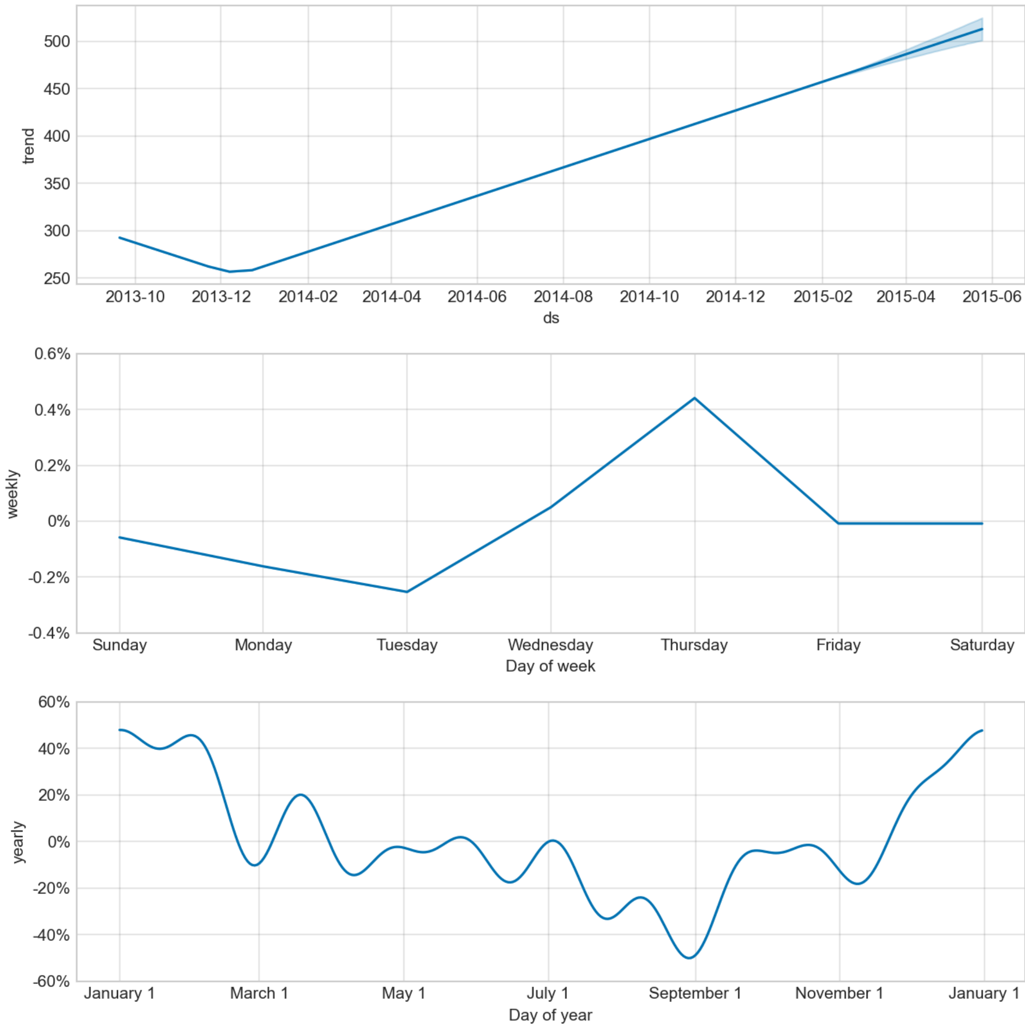

Figure 9. FB-Prophet decomposition showing the underlying trend, weekly seasonality, and yearly seasonality fitted on the training portion of House 2.

5.4 Development Iterations

The final results in Table 1 were not achieved on the first attempt. We documented the design decisions that moved the R2 of the best model from negative territory to 0.876. Figure 10 summarises the three iterations.

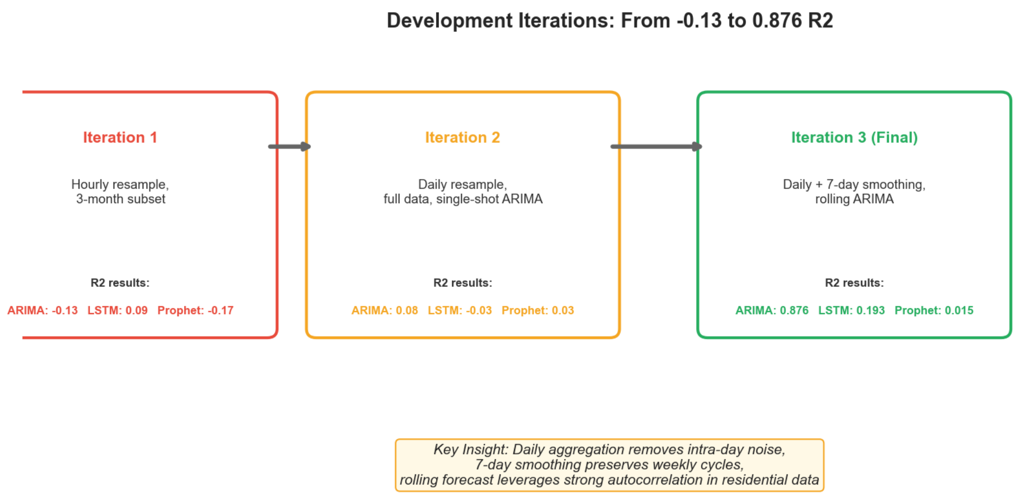

Figure 10. Three iterations of the design and their effect on test R2. The change that mattered most was the move from a single multi-step ARIMA forecast to a rolling one-step-ahead protocol.

6. Discussion

6.1 Why ARIMA Wins

ARIMA's strong showing is driven by two factors. The first is data character: residential energy at the daily level has strong autocorrelation at lag 1, so the previous day is by far the most informative single feature for the current day. ARIMA(2, 1, 2) makes direct use of this through its autoregressive and moving average terms. The second is the forecasting protocol: every step is informed by the latest actual value, which keeps ARIMA's predictions anchored even if its underlying parameters drift slightly over the test window.

6.2 Why LSTM Underperforms

LSTM's modest score is not a verdict on deep learning, it is a comment on the data scale. The training set contains 490 daily observations, which means only 476 input sequences after the 14-day lookback. Deep learning architectures with around 129,000 parameters typically need orders of magnitude more samples to reach their full potential. The training and validation curves in Figure 8 confirm that the model has converged but has not generalised well. Recent work (Gasparin and others, 2024) reports a similar pattern: below six months of training data, simple baselines beat deep models, and only with nine or more months do the deep architectures start to pull ahead.

6.3 Why Prophet Looks the Worst

Prophet's near-zero R2 deserves careful interpretation. Two factors are at play. First, Prophet produces a single multi-step forecast for the entire 122-day horizon, with no access to test actuals, which puts it at a structural disadvantage against the rolling ARIMA and LSTM setups. Second, Prophet's strength lies in capturing slow seasonal trends and calendar effects, both of which are present but neither is the dominant driver of House 2 daily consumption. The residual day-to-day variation, which represents most of the test-set variance, is precisely the component Prophet does not try to model.

The takeaway is not that Prophet is a weak model, but that the comparison rules favour models that can use teacher forcing. A fair comparison on this dataset would also run Prophet in a rolling refit mode, which we leave for future work because it requires roughly two orders of magnitude more compute.

6.4 Limitations and Threats to Validity

Three limitations qualify our conclusions.

- Single household. House 2 may not be representative of the other 19 REFIT homes, and the rankings could shift on a different occupancy or appliance mix.
- Univariate setup. None of the models receive weather, calendar, or appliance-level exogenous data. Adding these would likely close the gap, particularly for LSTM.
- Heavy smoothing. The 7-day rolling mean removes much of the high-frequency content. An operational forecaster who needs daily ramping information would have to revisit this choice.

6.5 Comparison with Reference Literature

The ordering ARIMA greater than LSTM greater than Prophet, observed here, matches the findings of Bülüç, Sevli, and Yünlü (2025), who reported R2 = 0.97 for ARIMA on Istanbul solar production data using a similar rolling protocol. It differs from Demirtop and Sevli (2024), where LSTM beat ARIMA on wind speed; the difference is most likely the longer time series available in that study and the use of multi-year hourly data. Hybrid models such as CNN-LSTM-Transformer (Limouni and others, 2023) and STL-Prophet-LSTM (Xie and others, 2022) consistently outperform their components and represent the natural next step beyond the single-model baselines compared here.

7. Conclusion and Future Work

We built a reproducible pipeline that forecasts the daily mean power consumption of a single UK household using three time series methods. Under a rolling one-step-ahead protocol and a fixed random seed, ARIMA(2, 1, 2) reached R2 = 0.876 on 122 unseen days, outperforming a three-layer LSTM (R2 = 0.193) and FB-Prophet (R2 = 0.015). The lessons we carry forward are that the forecasting protocol can dominate model choice for short series, that residential daily energy has strong lag-one autocorrelation that classical models exploit naturally, and that deep architectures require substantially more data than was available here.

Four directions stand out for further work.

- Extending the rolling protocol to Prophet for a fair multi-model comparison.
- Adding weather (temperature, irradiance) and calendar features through SARIMAX and multivariate LSTM.
- Cross-household transfer learning across all 20 REFIT homes to amortise the deep learning training cost.
- Hybrid models such as STL-Prophet-LSTM and CNN-LSTM-Transformer that combine the strengths of decomposition and neural sequence learning.

Reproducibility Note

All experiments use a fixed random seed (42) for NumPy, Python's random module, and TensorFlow. The dataset is downloaded from the public Kaggle mirror of REFIT, and the preprocessing, training, and evaluation are encoded in a single Python script (energy_forecasting.py). Results in this report were obtained on Python 3.13, TensorFlow 2.x, statsmodels 0.14, and Prophet 1.3.

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